# COMP2001: ARTIFICIAL INTELLIGENCE METHODS – LAB 1 EXERCISES

## RECOMMENDED TIMESCALES

This lab exercise is designed to take no longer than one week.

## GETTING SUPPORT

The computing exercises have been designed to facilitate flexible studying and can be worked through in your own time or during timetabled lab hours. It is recommended that you attend the labs, even if you have completed the exercises in your own time, to discuss your solutions and understanding with one of the lab support team and/or your peers. It is entirely possible that you might make a mistake in your solution which leads to a false observation and understanding.

## OBJECTIVES

The purpose of this lab exercise is to get set up with Java, IntelliJ IDE, and the COMP2001 Framework, and then use the COMP2001 framework to experiment with pseudorandom number generators for searching for solutions to an instance of the MAX-SAT problem. You will be using these systems in the subsequent lab exercises which together aims to give you the knowledge, understanding, and practical skills required to meet the learning outcomes of the computing part of the module. These include designing and implementing low-level operators, heuristics, and meta-heuristics for solving combinatorial optimisation problems. These learning outcomes will be the focus of the first in-lab assessment, so it is essential that you engage with these exercises.

The topics that will be covered in this exercise are:

- COMP2001 Framework.
- Maximum Satisfiability problem.
- Pseudo-random number generators.

## LEARNING OUTCOMES

By completing this lab exercise, you should meet the following learning outcomes:

- Students should be able to understand the Maximum Satisfiability (MAX-SAT) Problem such that they can explain how heuristics modify the solutions to MAX-SAT problems.
- Students should gain a practical understanding through experimentation of how pseudorandom number generators work and their consequences for computational experiments.
- Students should gain a practical insight into the COMP2001 Framework API to enable them to implement higher-level heuristic optimisation methods in future exercises.

## TASK 1 – GETTING SET UP WITH JAVA, INTELLIJ, AND THE COMP2001 FRAMEWORK

In this first task, we will guide you through setting up the software needed for completing the lab exercises. Depending on whether you plan on using the machines in A32, the computer science remote desktop, or your own machine, some of these may already be installed. If you are using your own device, please read the following "Java" instructions carefully as these may affect compatibility of your code between devices – *this is especially important for the project coursework*.

### JAVA

Java should already be installed on university devices; you can check this by opening a command line and entering the command "java -version". Many of you should already have installed Java from previous modules, however I highly recommend that you install the same Java version as those installed on the University devices to avoid compatibility issues. The following instructions are given for Windows, your experience on other devices may differ.

#### DOWNLOAD AND INSTALLATION INSTRUCTIONS

1. To download Java, go to the Latest Releases | Adoptium (link here) and download the ZIP archive for Java version 21.
2. Locate the archive and decompress it to get the corresponding folder which at the time of writing this document was *jdk-21.0.9+10*.
3. Move the entire jdk folder to a sensible location, for example "C:\Program Files\Java" if you are using Windows.
4. Now we need to tell the operating system where the Java binary resides by adding the folder to the system PATH. The following instructions is an example of how to do this on Windows.
    a. On Windows you can do this by opening the start menu and typing path. You should then see the option "Edit the system environment variables". Click on this and you should see a new window called "System Properties".
    b. Click on the "Environment Variables..." button (under the "Advanced" tab).
    c. In the new window look for the entry "Path" under "System variables", double click on the entry to open the "Edit environment variable" window.
    d. Press the "New" button" and enter the path to the directory containing the Java executable. If you put the JDK folder into "C:\Program Files\Java" then this should be "C:\Program Files\Java\jdk-21.0.9+10\bin".
    e. Press "OK" on all the open windows to close them.
    f. Check the operating system can find Java by opening a command line and entering "java -version". You should see the text **openjdk version "21.0.9" 2025-10-21 LTS**. If you do not, you may need to restart your device.

### INTELLIJ

#### DOWNLOAD INSTRUCTIONS.

IntelliJ (IntelliJ IDEA – the Leading Java and Kotlin IDE (jetbrains.com)) has two versions, a free to use community version, and a paid-for ultimate version. The ultimate version is free for students and can be obtained by making an account with your university email address. You can do this at the following link Free Educational Licenses - Community Support (jetbrains.com). Once set up, you should download the version of IntelliJ IDEA Ultimate that corresponds to your operating system and device architecture.

Alternatively, if you do not wish to create an account with JetBrains, you can download the free community edition at the following link https://www.jetbrains.com/idea/download/. Scroll down to find the community edition section.

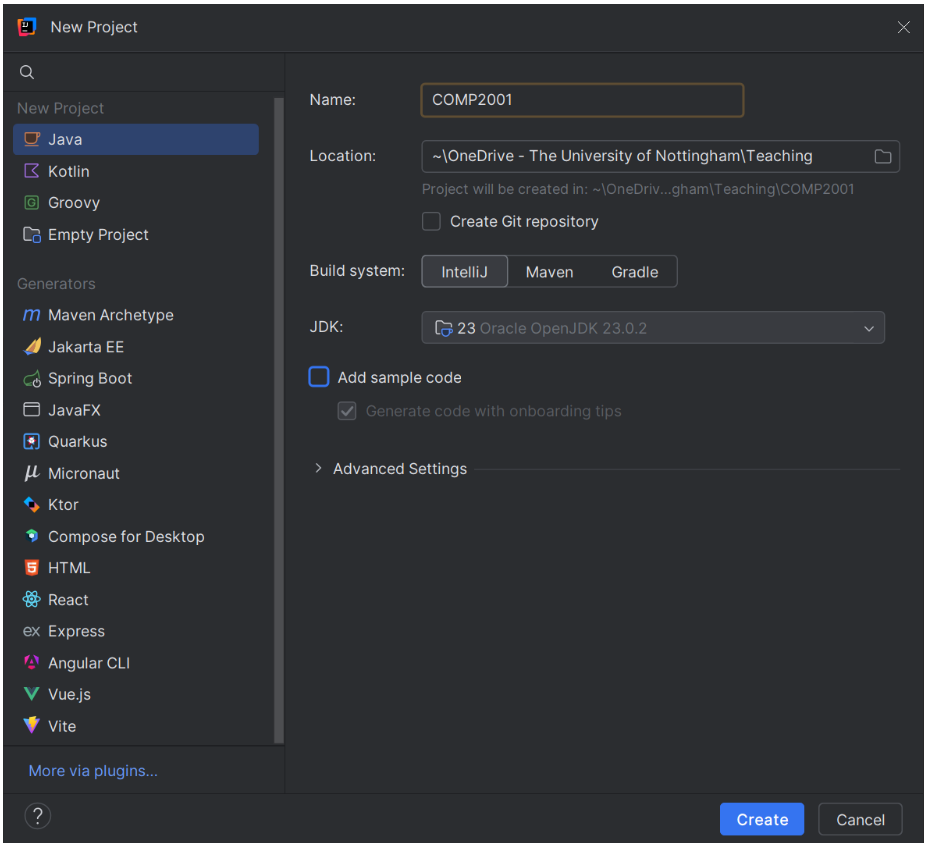
## INSTALLATION INSTRUCTIONS

**Note that this process may be different for non-windows operating systems.** To install IntelliJ and link the Java installation we installed in the previous step, run the installation package and install to your desired location. Choose “Next” on each page and finally press “Install”. You should not need to change any options (unless you wish to).

## COMP2001 FRAMEWORK

The COMP2001 Framework consists of a JAR file which can be downloaded from the module’s Moodle page along with the xchart dependency; search for and download “COMP2001 Framework JARs”.

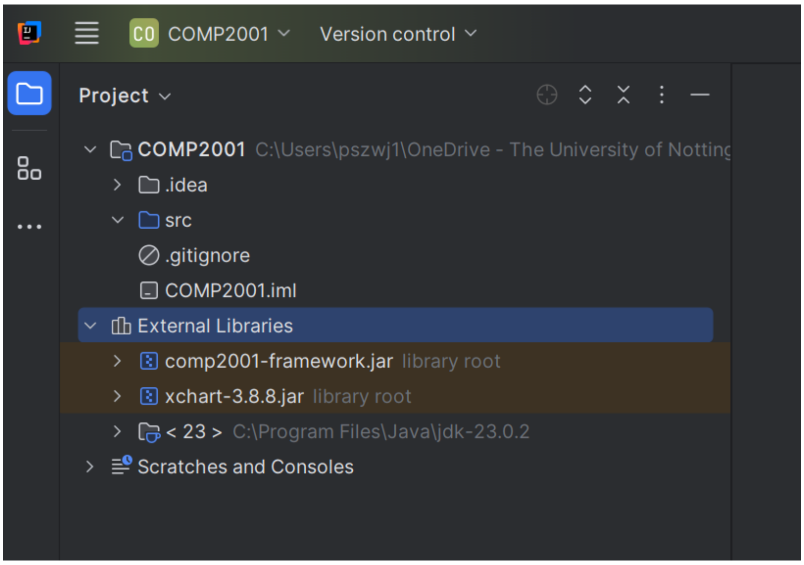
Open IntelliJ and choose to create a New Project. IntelliJ should automatically pick up the Java installation from the previous step. We can see this in the JDK dropdown. Name your project COMP2001 and make sure that it will be saved in a location which is persistent and backed up. We cannot be held responsible for loss of work if you have not backed it up using the university OneDrive, nor if you save it on a public device which is periodically wiped for security reasons (e.g. the lab machine C: drive). Uncheck “Add sample code” and press “Create”.




Now to include the COMP2001 Framework, we need to configure the project to locate the JAR files downloaded from Moodle. To do this, press on the button that has four horizontal lines in the top left of the window and choose “project structure”.

In the new window, select “Modules” on the left of the window, highlight the project (e.g. COMP2001), select "Dependencies”, and press on the + icon. Select “JARs or Directories…” and locate the folder containing the JAR files downloaded from Moodle. Select OK to get back to the main interface.

You can verify that this was successful by expanding “External Libraries” to see comp2001-framework.jar and xchart-3.8.8.jar.

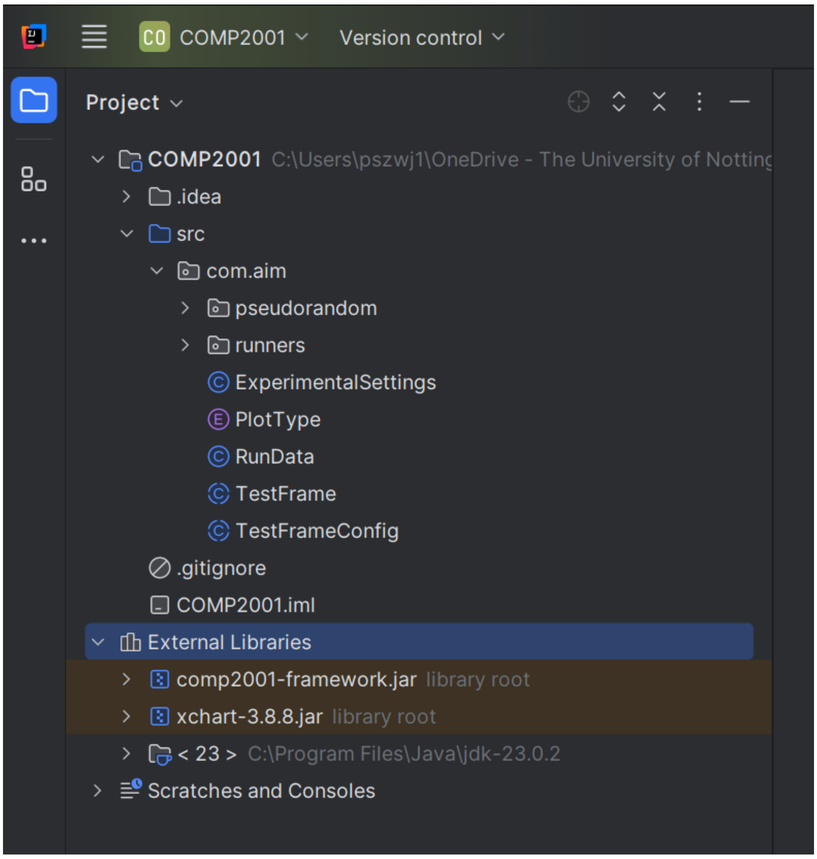



## TASK 2 – IMPORTING LAB EXERCISE FILES

Each lab exercise will introduce new functionality and requires different experiment runners, source files, and code templates. Each lab exercise code will depend on some of the files from this lab exercise, so it is important that you complete this exercise. Go ahead and download "Lab Exercise 1 Source Files".

Decompress the archive and locate the "com" folder inside the folder "src". Drag the "com" folder onto the "src" folder within IntelliJ to move the files into your project. IntelliJ might ask you to confirm the directory to move the files to; you can press on "Refactor" to proceed with the move. Your project structure should now look the same as the below screenshot.

You will need to follow the same set of steps for future lab exercises so you should remember what you did in this step.




## TASK 3 – THE MAXIMUM SATISFIABILITY PROBLEM

The maximum satisfiability problem (MAX-SAT) is an NP-hard optimisation problem and was the first of such problems to be proven so. A MAX-SAT problem instance is defined in terms of a number of Boolean variables and a number of disjunctive clauses connected by conjunction. This is known as conjunctive normal form (CNF). In this problem, we need to find an assignment of truths to each of the variables which maximises the total number of satisfiable clauses.

### OBJECTIVE FUNCTION

$\max z = f(X)$

Where:

$X$ is the vector of Boolean variables.

$f(X)$ counts the total number of satisfied clauses according to the variable assignments in $X$.

### PROBLEM INSTANCES

Here is an example of a MAX-SAT problem instance:

$$(A \lor \neg B) \land (B \lor C) \land (\neg A \lor \neg C)$$

Here, A, B, and C are the variables and can be either true or false. There are three clauses, $(A \lor \neg B)$, $(B \lor C)$, and $(\neg A \lor \neg C)$ which are all disjunctions of variables. If a variable has a truth assignment in one clause, then that same variable has the same truth assignment across all clauses. In the MAX-SAT problem, the goal is to find a truth assignment to A, B, and C which maximises the number of satisfied clauses. In this example, we can assign $A = true$ so satisfy the first clause, $B = true$ to satisfy the second clause, and for the third clause, we cannot reassign A to be false otherwise the first clause becomes UNSAT. We therefore need to assign $C = false$ to satisfy the third clause.

## SOLUTION REPRESENTATION

Typically, and within the COMP2001 framework, a solution to a MAX-SAT problem can be represented as a binary string (a sequence of bits which are true or false) where each bit represents the truth value of each variable. In this example, an optimal solution can be represented as "110" assigning $A = True, B = True, C = False$.

## TASK 4 – FORMATIVE ASSESSMENT PART 1

On the Moodle page you should find a link to "Lab 1 Formative Assessment". The first two questions are to allow you to test your understanding of the Maximum Satisfiability problem. These questions are multiple choice questions and should get instant feedback. If you wish to discuss your answers or are not quite sure of the correct answer, find one of the lab assistants or convenors who can discuss this with you.

## TASK 5 – THE BITFLIP OPERATOR AND PSEUDORANDOM NUMBER GENERATORS

Please ensure that you have completed task 4 and agree with your answers and feedback before attempting this task as it is important that you understand the problem we are solving and its representation.

## FAMILIARISATION WITH THE SOURCE CODES

Going back to IntelliJ (or your preferred IDE), you should see that there are a number of source codes in the package **com.aim.pseudorandom**. The practical aspect of this lab exercise is centred around these source files, and an exercise specific exercise runner class.

Each lab exercise will have its own runner class(es) and will be numbered corresponding to the lab exercise number. You will receive new runner classes for each exercise and you can find these files in the package **com.aim.runners**. Each exercise will contain a pre-defined runnable test frame class (in this case Lab1ExercisesRunner.java) and a configuration (in this case Lab1ExercisesTestFrameConfig) which allows you to change some settings within the experimentation; you will be instructed which settings you should change in the corresponding exercise sheet.

## RANDOMISED SEARCH

The test frame in this exercise runs a random walk search method which iteratively randomly modifies the solution-in-hand. Open the **RandomWalk** class and let's see what is happening. Within the run method, there is some code to keep track of actual time taken. This is not very interesting but the code that the timing encapsulates is the important part. Here we are initialising a new solution and iteratively flipping random bits in the bitstring that represents the solution to the problem being solved. Look at each line of code and the corresponding comment. Refer to the COMP2001 API document to understand what each line of code is doing.

## EXPERIMENT RUNNERS AND TERMINATION CRITERION

Now open the experiment runner Lab1ExercisesRunner and try running the test frame for this lab exercise – what happens? You should observe that the program never terminates. This is due to the termination criterion of the experiments, and because we have not implemented the randomised bitflip low-level operator. Open the RandomBitFlipHeuristic.java file. You will see a method with the signature "`public void applyHeuristic(SAT oProblem, int iSolutionIndex)`" which implements an abstract method in its parent SATHeuristic class. At the moment this does nothing which means the solution-in-hand is never modified. For fairness and repeatability of experiments across different devices, the termination criterion within the COMP2001 framework is implemented as a ratio between the number of operations achieved in 10 nominal minutes on a reference machine and the number of evaluations of solutions that have been modified since they were last evaluated. Since we didn't modify the solution before evaluating the objective value of the solution, we considered this to not contribute to the search and hence we never reach the limit of the termination criterion.

## RANDOM BIT FLIP HEURISTIC

To implement a randomised bit flip heuristic, we need to understand a few methods in the API. Go ahead and look up the following in the API document that can be found on the Moodle page under the section "Computing Sessions (COMP2001)" before continuing:

- getNumberOfVariables() in the SAT class.
- bitFlip(int iBitIndex, int iMemoryIndex) in the SAT class.

Within this heuristic, we want to be able to choose one of the variables in the solution at random and change its truth value. Try to implement this yourself using the API methods above as a hint. You can use Java's Random class to generate a random integer. **Note that a seeded random number generator is already defined in the SATHeuristic class with the variable name m_oRandom.**

After you have completed the implementation, try running the test frame again. If you implemented it correctly (and haven't modified any configurations) you should get an output with shows the results of running three different trials with the same seed value to get a solution which has an objective value of **364**. If your solution is different, you should discuss your implementation with your peers ask one of the lab assistants in the lab for help.

## PSEUDORANDOM NUMBER GENERATORS

At the end of the previous exercise, you should have run the test frame which executes an experiment consisting of three independent trials using the seed `30012026L`. This is the value that is used to seed the random number generator [Random (Java SE 21 & JDK 21) (link to Javadocs here)](#).

## TASK 6 – FORMATIVE ASSESSMENT PART 2

The remaining four questions in the “Lab 1 Formative Assessment” are a mix of multiple-choice questions and open textbox questions. Answer the remaining questions in combination by modifying the code as appropriate.

**Note that** feedback for open textbox questions will be provided no later than the next lab and will address common themes at the cohort level. If you wish to discuss individual answers, please ask one of us during one of the lab sessions.